End-to-End Experimental Workflow: AST-Driven Multi-Paradigm OLAP Benchmarking (75 Queries, 6 Engines)

PHASE 1: TPC-H Data Generation & Multi-Layout Storage (SF = 1 GB)

- Base Dataset: 8 Relational Tables (8.66M rows; lineitem = 6.00M rows)
- Local Columnar: ZSTD Parquet + DuckDB Native + Apache Arrow Buffers
- Local Row-Store Baseline: SQLite 3 In-Memory B-Tree with PK/FK Indexes
- Cloud Serverless: Google Cloud Storage -> BigQuery Unoptimized vs. Clustered

PHASE 2: 75-Query Factorial Workload Design (25 Seeds x 3 Variations)

- 5 Structural Tiers (5 Canonical Seeds/Tier = 25 Independent Seeds):
  L1 (Scan/Filter), L2 (4-8 Way Joins), L3 (Staged CTEs), L4 (Window), L5 (UNNEST)
- 3 Selectivity Perturbations per Seed (Total N = 75 Queries):
  V1: High Selectivity (1-5%) | V2: Medium (15-30%) | V3: Full-Horizon (65-95%)

PHASE 3: Static Abstract Syntax Tree (AST) Profiling & Complexity Quantification (SQLGlot Parser)

- Parses all 75 SQL queries into canonical AST representations to extract structural operator vectors:
  Tree Depth D(q), Total Operator Nodes N(q), Join Count J(q), Nested CTE/Subquery Depth S(q), Window Specs W(q), and Array UNNEST U(q)
- Composite Structural Complexity Index: $CSCI(q) = 1.5\,D(q) + 2.5\,J(q) + 3.0\,S(q) + 3.5\,W(q) + 4.0\,U(q) + 1.5\,C(q)$ [Range: 21.0 (L1) to 36.4 (L5)]

Paradigm 1: Row-Store

- Engine: SQLite 3 (ANSI C)
- Storage: In-Memory B-Tree
- Execution: Tuple-at-a-Time Volcano Iterator Model
- Role: Traditional Row-Store Baseline (Mean: 5071.1 ms)

Paradigm 2: Serverless Cloud

- Engine: Google BigQuery
- Storage: Capacitor Columnar (Unoptimized vs. Clustered)
- Controls: Cache Disabled, Zero-Billing Sandbox Quota
- Telemetry: Wall & Slot Time, Bytes Scanned (745.4 ms)

Paradigm 3: Vectorized CPU

- 1) DuckDB (C++ Columnar) Parquet vs. Native (183.7ms)
- 2) Polars SQL (Rust Arrow) Lazy Streaming (66.5 ms)
- 3) Apache DataFusion (Rust) Arrow Planner (88.2 ms)

Paradigm 4: SIMT CUDA GPU

- Engine: PyTorch CUDA OLAP
- GPU: NVIDIA RTX 2050 (4GB)
- Hardware cudaEvent Timers: T_Cold = T_PCIe_H2D + T_Warm
- Mean PCIe H2D: 34.71 ms
- Mean Warm Kernel: 22.25 ms

PHASE 5: Statistical Verification & Efficiency Metrics

- Result-Set Multiset Equivalence Check: EX(q) in {0, 1}
- Valid Efficiency Score (VES) & Storage Pruning Ratio (78.4% in V1)
- 10,000-Iteration Seed-Cluster Bootstrap 95% CIs (N_seeds = 25)
- Paired Two-Sided Wilcoxon Signed-Rank Tests (Holm-Bonferroni)
- Architectural Crossover Analysis (L1 PCIe H2D vs. L4/L5 Speedup)

PHASE 6: Cryptographic Audit & Manuscript Artifacts

- Automated LaTeX Booktabs Tables (Tables I-V) & 300 DPI Figures
- Zero-Discrepancy Verification Suite (08_verify_paper_statistics.py)
- SHA-256 Telemetry Audit Certificate (REPRODUCTION_CERTIFICATE.md)
- Complete Open-Source Benchmark Repository & IEEE Access Manuscript:
  https://github.com/preecha-msu/bigquery-olap-benchmark